PromoProof: Source-Grounded Promo Discovery

Continuity-Aware Ad Shortlist Generation for Microdrama Screenplays | Problem 1

Dipanshu Raj • **Roll No: 230375**
IIT Kanpur • Kuku Technologies AI Generalist Placement
Problem 1 Submission • Production Prototype Verified

1. PROBLEM CHOICE, FRAMING & STRATEGIC ASSUMPTIONS

Why Problem 1: Acquisition editors at Kuku Technologies manually scrub dozens of 1–2 minute microdrama episodes weekly to select 20–60s ad clips for TikTok, Instagram Reels, and YouTube Shorts. Problem 1 provides the highest bounded leverage: both inputs (structured screenplay packs) and editorial goals (paid audience acquisition) are fully defined. A script linter requires downstream video assets, and a paywall optimizer demands behavioral streaming metrics absent in text. PromoProof focuses strictly on **traceable, defensible discovery** rather than ungrounded claims of creative virality.

Core Operational Assumptions:

- **Ad Creative Physics:** Scroll-stopping ads require a shock hook within the first 3s (slap, revelation, confrontation) and must cut on an urgent, unresolved cliffhanger.
- **Dialogue Safety Mandate:** Zero mid-sentence cuts. A boundary t is strictly illegal if an actor is speaking: $start < t < end$ for any DIALOGUE block.
- **Shortlist vs. Dump:** An editor needs the top 12–16 high-conviction ad candidates across 50 episodes—not 50 mediocre summaries. Discarding weak episodes is essential.

Design Principles & Non-Goals:

- **Deterministic Boundaries:** Code owns integer-millisecond math, legal timestamps, and deduplication; LLMs cannot hallucinate cut points or invent timestamps.
- **Zero Data Retention (ZDR):** Confidential show IP is processed in volatile memory only; never stored on disk or local databases, and never used for LLM training.
- **Explicit Non-Goals:** No raw video rendering, no auto-captioning, and no uncalibrated CTR/ROAS conversion forecasts.

2. SYSTEM ARCHITECTURE & RESPONSIBILITY PARTITION

PromoProof implements a 4-tier pipeline: **Browser SPA (React 18 / Tailwind)** → **In-Memory Express Server (Node 22)** → **Python Analytics Engine (engine.py)** → **Remote LPU Inference (Groq gpt-oss-120b).** Responsibility is strictly partitioned:

Pipeline Stage	Deterministic Code Owns (Pure Python)	LLM Owns (Groq LPU / OpenAI Adapter)	Human Review Gate
1. Ingest & Parse	ZIP path traversal security, integer-ms timing, scene intervals, block graph, SHA256 hashes.	*Nothing (preserves context; prevents prompt hallucination & token waste).*	Inspects malformed headers in Run Audit.
2. Retrieval	Scans all safe 20–60s windows; clusters acoustic shocks within 1.5s; 55% overlap dedup.	*Nothing (retrieves top 16 candidates deterministically).*	Adjusts duration & audience filters.
3. Semantic Pass	Compact payload (strips middle camera descriptions; 86.7% token reduction); bounded ID repair.	Scores cold-open coherence, stakes, and cliffhanger urgency (0–4 integers).	Reviews narrative rationale & risk flags.
4. Critic & Gate	**Deterministic Gate:** enforces scores ≥3, hook ≤3s, cliff ≥last 5s, non-empty open question.	Adversarial critic challenges top 5 candidates for pronoun ambiguity & resolved endings.	**Final Accept / Reject decision + notes.**

3. CONTINUITY DESK, MODEL ADAPTER & CONFIDENTIALITY

Entity Drift & Scoped Identity: In microdramas, male leads evolve across episodes (e.g. THE BEARDED STRANGER → SCARRED BEARDED MAN → ALEXANDER), while generic titles (e.g. BLOND NOBLEMAN) recur in Episodes 2, 9, 27, and 44 representing completely different background actors. Rather than attempting dangerous automated global alias merges—which spoil reveals or merge distinct characters—the **Continuity Desk** normalizes text, flags ambiguous identities, and scopes them to individual episodes as reviewer annotations.

Modular Model Adapter: engine.py implements a unified OpenAI-compatible adapter. Switching between Groq (openai/gpt-oss-120b), OpenAI (gpt-4o), or self-hosted vLLM requires changing two server variables (OPENAI_API_KEY, OPENAI_BASE_URL) with zero parser modifications. Includes 6-attempt exponential backoff respecting HTTP 429 Retry-After headers. Scripts stream through volatile memory only; HTTP logs omit request/response bodies to safeguard unreleased IP.

PromoProof: Evidence, Benchmark & Quality Audit

50-Episode Season Verification, Honest Failure Analysis & Scale Economics

Dipanshu Raj • **Roll No: 230375**
IIT Kanpur • Problem 1 Verification Report
Status: **Verified Production Deployment**

4. MEASURED FULL-PACK BENCHMARK (50 EPISODES — "I MARRIED A BROKEN SOLDIER")

The complete 50-episode screenplay pack (529 KB, 3,727 timestamped blocks, 169k raw tokens) was ingested and audited in memory:

50 / 50 EPISODES AUDITED	**98,309** SAFE WINDOWS SCANNED	**16** SHORTLISTED PROPOSALS	**86.74%** TOKEN REDUCTION	**$0.0099** FULL SEASON INFERENCE	**46 / 46** UNIT TESTS PASSED

5. CANDIDATE RECOMMENDATIONS: WHAT TO POST FIRST & WHY

ID	Episode & Timing	Duration	First 3s Opening Hook	Closing Cliffhanger	Gate Status	Funnel Priority
P001	**EP20** 00:15.4–01:09.1	53.7s (Cont)	Catherine raises porcelain teacup with sharp, dismissive glare.	*"If I stop, Seamus, the lung fever in the valley wins."*	**PASSED (1/16)**	Broad Cold #1
P007	**EP09** 00:52.5–01:41.9	49.4s (Cont)	Mother's desperate cry: *"Why isn't it working?"*	Nobleman demands: *"Where did you get this?"*	**Review Priority**	Broad Cold #2
P002	**EP28** 00:30.3–01:11.2	40.9s (Cont)	Duchess accuses from bed: poison accusation.	Catherine violently yanks silver locket off Juliana's neck.	**Review Risk**	Broad Cold #3
P016	**EP47** 00:42.5–01:24.0	35.8s (Comp)	Alexander locks gaze onto Juliana's tearful eyes.	Alexander turns in shock as Catherine flees.	**FAILED GATE**	Trope Affinity

6. HONEST FAILURE MODES & THE P016 VALIDATOR DISCOVERY

In accordance with the assignment brief's core requirement to disclose where the tool fails and where AI got it wrong:

The P016 Validator Loophole (Caught & Fixed):

In our initial live Groq run, candidate **P016** (Episode 47 composite) received a pass from the model despite having a stitch_legibility score of only 1/4 and a blank open_question. The original validator naively trusted the top-level verdict. We tightened deterministic code gates in apply_reviews() to mandate sub-scores ≥3 and explicit open questions. Replaying recorded outputs correctly failed P016 with STITCH_GATE_FAILED and OPEN_QUESTION_MISSING, preserving exactly 1 high-confidence pass (P001).

Acoustic False-Positives & Lore Blindness:

Carriage-Door Bug (Ep 7): An early baseline heuristic double-counted a carriage door slamming in both ACTION and SOUND FX as two distinct hooks, failing to recognize that the scene ended on a resolved smile. We fixed this by clustering acoustic impacts within 1.5s.

Lore-Dependent Cliffhangers (P002): The Duchess accuses 'her' of poisoning, but cold viewers cannot tell who is speaking without prior backstory (pronoun ambiguity).

7. ECONOMICS, SCALE TO 100 SHOWS/WEEK & PRODUCTION ROADMAP

• **Measured Ingest Economics:** 50 episodes required 37,365 input and 7,215 output tokens across 11 Groq requests ($0.00993 per show). Scaling to 100 shows/week (5,000 episodes) costs **$0.99/week** in inference.
• **Production Architecture:** Decouple Express with Redis Queue / Celery workers. Stateless Python processes parse episodes in 10ms each, running on basic shared CPU compute without GPU instances.
• **Multimodal Video Roadmap:** (1) Align screenplay timestamps to MP4 audio using word-level Whisper. (2) Computer vision (YOLO/MediaPipe) to verify actor facial expressions and shot density in the first 3s. (3) Automated 9:16 vertical crop with burned-in kinetic captions.

8. VERIFICATION STATUS & PROTOTYPE ACCESS

Automated Verification: 46 unit tests passing (36 engine + 10 deployment safety); GitHub Actions CI green; Docker container smoke test verified.
Live Web Prototype: https://promoproof-bvhz.onrender.com/#/ (Active Groq AI connected; live in-browser testing).
GitHub Repository: https://github.com/Dipanshur19/promoproof (Private repository; collaborator access granted upon request).
Raw Reviewer Artifacts: Attached PromoProof Confidential Reviewer Outputs.zip containing 17 unedited JSON/CSV execution logs.